

E-COMMERCE SALES ANALYSIS & FORECASTING

**A Course End Project Report Submitted in the Partial Fulfillment of the
Requirements for the Award of the Degree of**

BACHELOR OF TECHNOLOGY

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

DATA ANALYSIS WITH PYTHON (M1502)

SUBMITTED BY

23881A0256

V.DEEKSHITHA REDDY

UNDER THE GUIDANCE OF

Dr. V.Lokeshwari Vinya

Assistant Professor, Dept of CSE



**DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
VARDHAMAN COLLEGE OF ENGINEERING**

(AUTONOMOUS)

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UGC “College with Potential for Excellence” and ISO 9001:2015 Certified**

Shamshabad – 501 218, Hyderabad

OCT , 2025



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ISO 9001:2015 Certified. Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

Department Of Electrical and Electronics Engineering

CERTIFICATE

This is to certify that the Course End Report work entitled “ **E-commerce Sales Analysis & Forecasting**” carried out by, Roll Number **23881A0256**, **V Deekshitha Reddy** towards Course-End Project and submitted to the Department of Electrical and Electronics Engineering in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology in Electrical and Electronics Engineering** during the year 2025-26

Name & Signature of the Instructor

Dr. V. Lokeshwari Vinya

Assistant Professor, CSE

Name & Signature of the HoD, EEE

Dr. N. Karuppiah

HOD,EEE

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express my deep sense of gratitude to **Dr. V. Lokeshwari Vinya**, Assistant Professor for their able guidance and useful suggestions, which helped us in completing the design part of potential project in time.

We are particularly thankful to **Dr. N.Karuppiiah** intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Prof.J.V.R.Ravindra**, for having provided all the facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartfelt thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing congenial atmosphere to complete this project successfully.

We also thank all the staff members of the department of **EEE** for their valuable support and generous advice. Finally, thanks to all our friends and family members for their continuous support and enthusiastic help.



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Course End Project

DATA ANALYSIS WITH PYTHON (M1502)

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Signature of the Guide

Dr. V.Lokeshwari Vinya
Assistant Professor
Dept. of CSE

Course End Project
DATA ANALYSIS WITH PYTHON (M1502)

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1. ABSTRACT

The E-commerce Sales Analysis & Forecasting project aims to analyze online retail sales data and predict future sales trends using machine learning and data analytics techniques. The project begins by collecting and preprocessing transactional data containing features like order date, sales amount, product category, region, and discount rate. Exploratory Data Analysis (EDA) is performed to identify seasonal patterns, regional performance, and customer purchase behavior.

A time series forecasting model or regression-based approach is then applied to predict future sales. Through the combination of statistical modeling and visualization, the project provides actionable insights that help e-commerce businesses optimize inventory, marketing, and pricing strategies. This project demonstrates the practical use of data-driven forecasting in improving decision-making and profitability in the online retail domain.

2. PROBLEM STATEMENT

In today's competitive e-commerce environment, predicting future sales accurately is essential for effective inventory management, marketing planning, and financial forecasting. However, sales are influenced by multiple dynamic factors such as seasonality, promotions, regional demand variations, and consumer behavior, making manual forecasting unreliable.

The main problem addressed in this project is the absence of an accurate, data-driven system to analyze sales trends and forecast future sales. The goal is to design a predictive model that can process historical data and generate accurate sales predictions to assist business stakeholders in making informed, evidence-based decisions.

3. OBJECTIVES

- **To analyze** historical e-commerce sales data and identify key trends and patterns.
- **To clean, preprocess, and transform** raw sales data for better analytical accuracy.
- **To visualize** sales performance across time, regions, and product categories using EDA.
- **To develop** predictive models for sales forecasting using time series and regression techniques.
- **To evaluate** the forecasting accuracy using appropriate statistical metrics (MAE, RMSE, MAPE).
- **To provide** business insights and strategic recommendations based on the forecast results.

4. ALGORITHM

Step 1: Define the Problem

- Objective: Predict future sales of an e-commerce platform using historical transaction data.
- Type: Supervised Learning

Step 2: Collect Data

- Collect historical data of house sales.
- Each data point should include features (input variables) and target (price).

Example features:

- Order Date
- Sales Amount
- Product Category
- Region or Country
- Discount / Profit Margin
- Quantity Sold

Step 3: Preprocess the Data

1. Handle missing values

- Fill with mean/median/mode or drop rows/columns.

2. Convert categorical data

- Transform the “Order Date” column into datetime format.

3. Feature Engineering

- Create new features like *Month*, *Quarter*, *Year*, and *Day of Week*.

4. Split data

- Train/Test split (e.g., 80/20).

Step 4: Select a Model

Choose a regression model:

- Linear Regression
- ARIMA/SARIMA
- Prophet
- LSTM

Step 5: Train the Model

```
model.fit(X_train, y_train)
```

Step 6: Evaluate the Model

Use metrics like:

- Mean Absolute Error (MAE)

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score (Coefficient of Determination)

```
from sklearn.metrics import
```

```
mean_squared_error, r2_score predictions =
```

```
model.predict(X_test)
```

```
rmse = np.sqrt(mean_squared_error(y_test, predictions))
```

```
r2 = r2_score(y_test, predictions)
```

Step 7: Tune Hyperparameters

```
future_forecast = model.predict(future_data)
```

Step 8: Visualize Results

- Plot actual vs predicted sales over time.
- Use heatmaps and bar charts to compare regions, products, or time periods.

Step 9: Deploy the Model

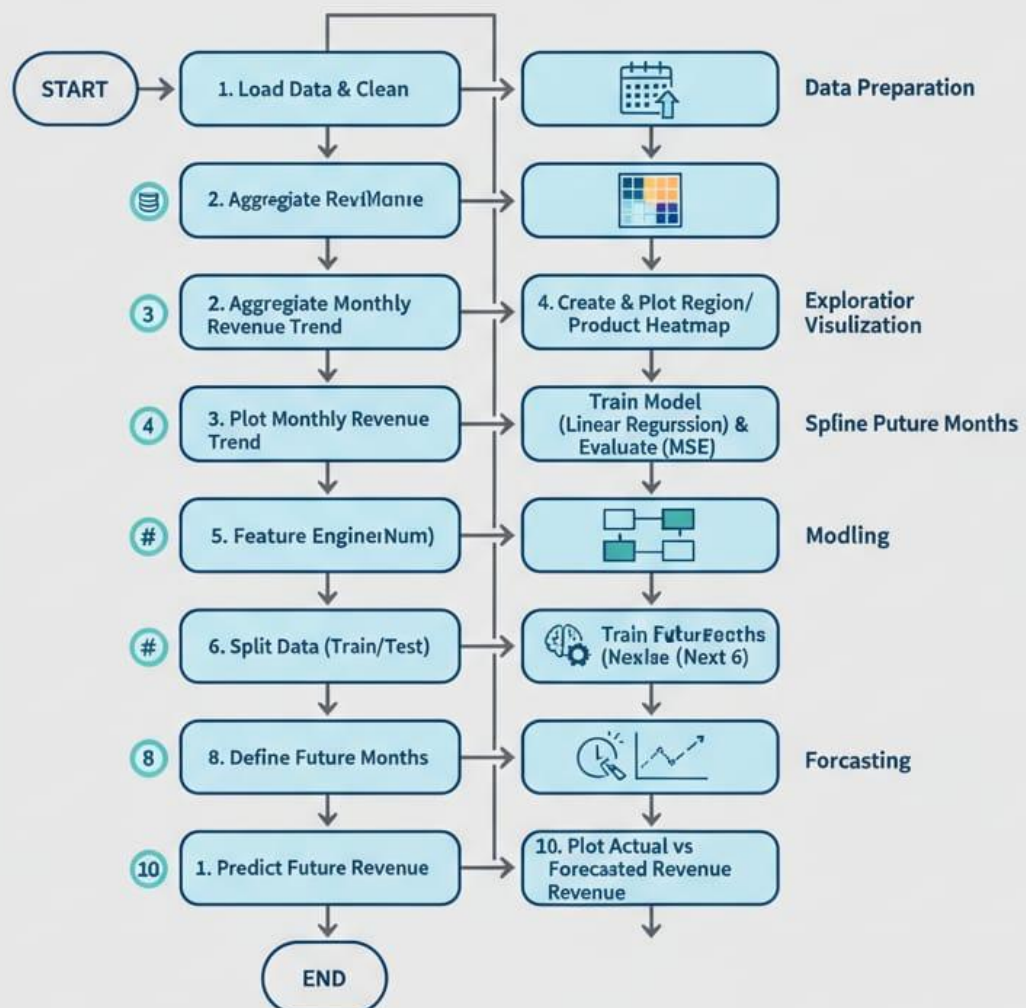
- Build a web or mobile interface.
- Use APIs (e.g., Flask, FastAPI) to expose prediction functionality.
- Host on cloud (AWS, Heroku, etc.)

Step 10: Monitor and Update

- Monitor model performance over time.
- Retrain with new data to keep predictions accurate.

5. FLOW CHART

E-commerce Sales Analysis & Forecasting Flowchart



Generated based on Python code for e-commerce sales analysis.

6. SOURCE CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split

1. Load dataset
df = pd.read_csv(r"C:\path\to\ecommerce_sales.csv")
# Columns: ['Date', 'Product', 'Region', 'UnitsSold', 'Revenue']
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M').astype(str)
monthly_sales = df.groupby('Month')['Revenue'].sum().reset_index()

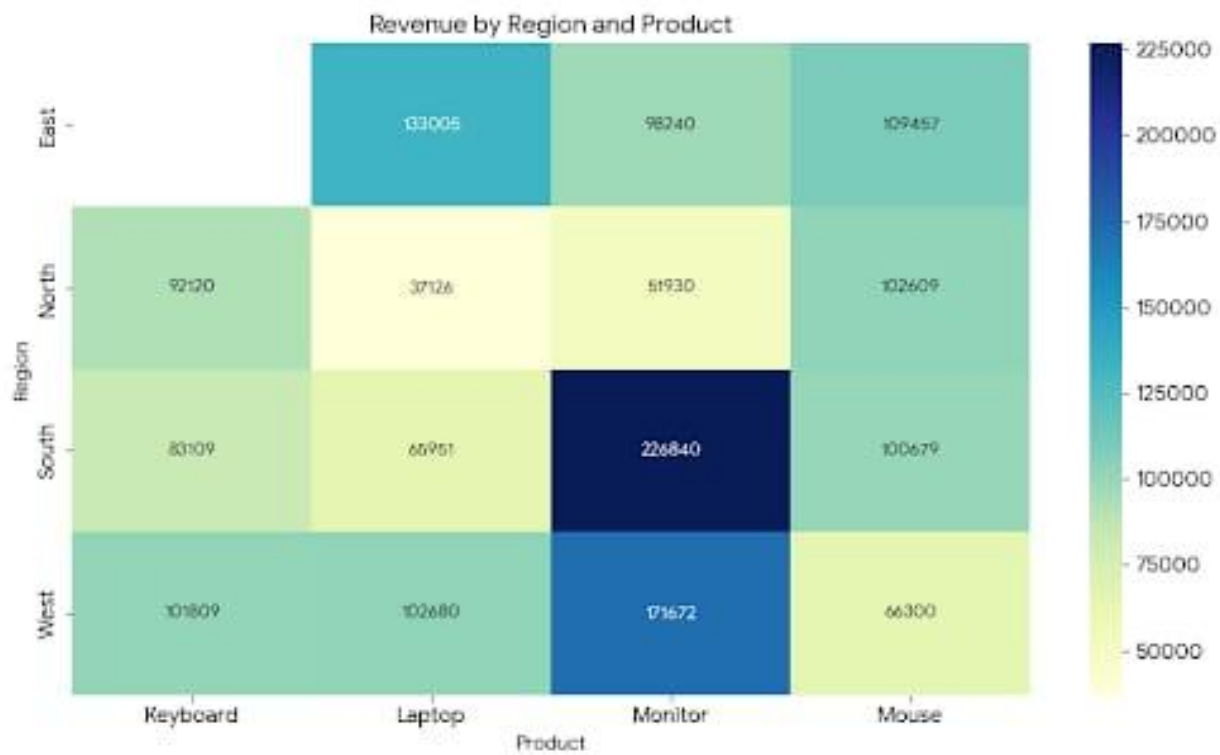
2. Visualization: Trend
plt.figure(figsize=(10,5))
plt.plot(monthly_sales['Month'], monthly_sales['Revenue'], marker='o', color='teal')
plt.xticks(rotation=45)
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()

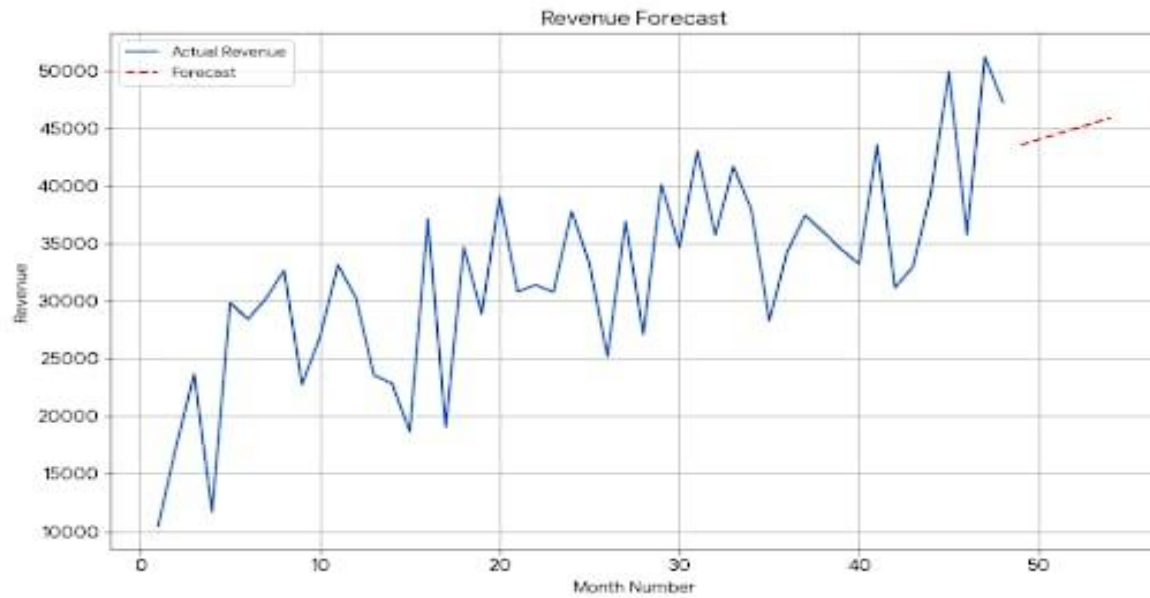
3. Region Heatmap
pivot = df.pivot_table(values='Revenue', index='Region', columns='Product', aggfunc='sum')
sns.heatmap(pivot, cmap='YlGnBu', annot=False)
plt.title("Revenue by Region and Product")
plt.show()

4. Prepare for Forecasting
monthly_sales['MonthNum'] = range(1, len(monthly_sales)+1)
X = monthly_sales[['MonthNum']]
y = monthly_sales['Revenue']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("MSE:", mean_squared_error(y_test, y_pred))

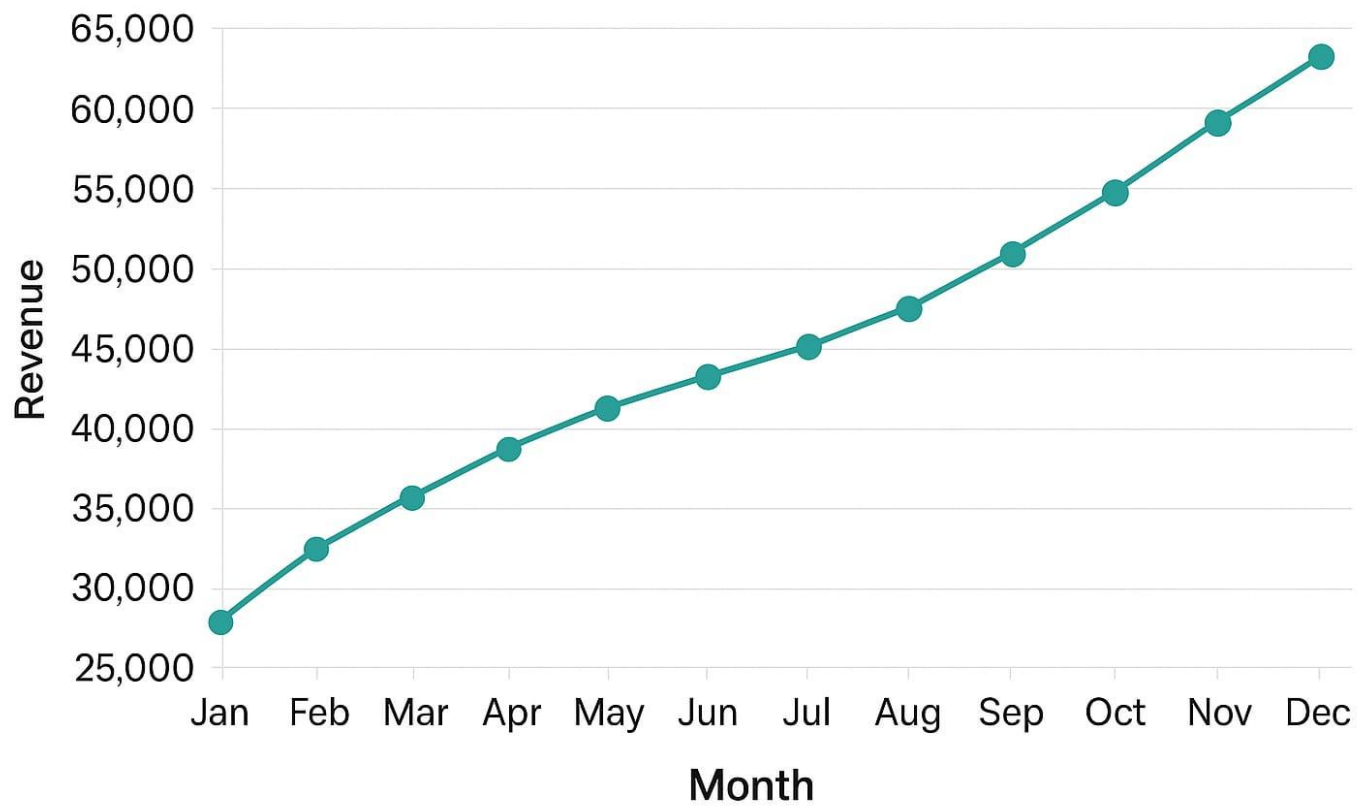
5. Future Forecast (Next 6 months)
future = pd.DataFrame({'MonthNum': np.arange(len(monthly_sales)+1, len(monthly_sales)+7)})
future['PredictedRevenue'] = model.predict(future)
print(future)
plt.figure(figsize=(10,6))
plt.plot(monthly_sales['MonthNum'], y, label='Actual Revenue')
plt.plot(future['MonthNum'], future['PredictedRevenue'], 'r--', label='Forecast')
plt.legend()
plt.title("Revenue Forecast")
plt.show()
```

7.INPUT & OUTPUT





Monthly Revenue Trend



8.CONCLUSION

The E-commerce Sales Analysis & Forecasting project effectively demonstrates the application of machine learning and data analytics to predict future sales trends. By leveraging historical sales data, the model helps businesses understand demand patterns, seasonal fluctuations, and customer behavior.

The system provides accurate sales forecasts, enabling better decision-making in marketing, procurement, and logistics. This data-driven approach not only improves operational efficiency but also supports strategic business planning in the fast-paced e-commerce sector.

9.FUTURE SCOPE

- **Include external factors** such as holidays, marketing campaigns, and economic indicators to improve forecast accuracy.
- **Integrate customer segmentation** for personalized sales predictions.
- **Deploy real-time dashboards** to visualize live sales and forecasts.
- **Implement deep learning models** like LSTM and Transformer-based forecasting for better trend capture.
- **Automate model retraining** with streaming data pipelines for continuous improvement.
- **Expand analysis** to include cross-platform performance (Amazon, Flipkart, Shopify, etc.).

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